

DATA ANALYSIS INTERNSHIP

Project Report Submitted

to

SWYNEX Technologies



by

Vinayak Triguna Pandey

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PROBLEM STATEMENT

In this project, there are 3 tasks we aim to address. In first task, I aim to address problem of choosing a public dataset and then cleaning it i.e. identifying missing values, duplicate records, incorrect data types and inconsistent values, with the help of excel, sql or python. In Task 2, I aim to address problem of performing exploratory data analysis on cleaned dataset from Task 1 using excel, sql or python and calculating important statistics and identifying trends and patterns. In Task 3, I aim to address problem of creating an interactive dashboard using Power BI, Tableau or another suitable BI tool.

DATASET INFORMATION

The dataset which I have used in this project is Titanic dataset. It has been downloaded from Kaggle. It has 891 rows and 12 columns – PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin and Embarked. This dataset was cleaned using python and pandas in Google Colab.

CLEANING PROCESS

First, the cleaned titanic dataset was uploaded to google colab online. Then duplicates were removed. Then, the name column was split into 2 columns – First_name and Last_name. Then data type of each column was checked. Everything was Ok there. Then, percentage of missing/null values was checked in each column. Some percentage was found in Age, Cabin and Embarked columns. Cabin column was deleted because more than 70% of data in this column was missing or had null values. Then, missing age rows was filled with median age. Then, missing embarked fields was filled using the mode. Finally, the dataset was downloaded from google colab and was edited manually so that it looks structured. Cleaned dataset will be shared in github repository.

ANALYSIS

After cleaning the titanic dataset, I performed exploratory data analysis on this dataset. I uploaded this dataset in google colab online. Then I did univariate analysis. It means that I plotted graph for overall survival count,passenger count by class and age distribution of passengers. Then, I did bivariate analysis by plotting graph for survival count by gender, survival count by passenger class and age distribution split by survival status. Then I did multivariate analysis by plotting graph for survival rate across class and gender. So, in total I plotted 7 graphs. Cleaned dataset and google colab file which when opened in google colab opens EDA python code performed on cleaned dataset will be shared in github repository.

DASHBOARD

The clean titanic dataset was uploaded to tableau public online. Three KPIs were created- Total Passengers, Survival Rate and Average Age and Average Fare. Then 3 charts were created- Survival by class, Survival by Gender and Age Distribution. Then these 6 are grouped together to form interactive dashboard. The tableau file with extension .twbx will be shared in github repository along with cleaned dataset.

KEY BUSINESS INSIGHTS

The key business insights I get after going through data analysis process on titanic dataset are:

i) 577 male passengers survived and 314 female passengers survived.

ii) Total passengers survived in Pclass 1- 216.

iii) Total passengers survived in Pclass 2-184.

iv) Total passengers survived in Pclass 3-491.

v) There were 891 passengers.

vi) Survival rate is 38.4%.

vii) Average age of a passenger is 29 and average fare is 32.20.

